

Nutritional quality of legumes, and their role in cardiometabolic risk prevention: a review.

Legumes (including alfalfa, clover, lupins, green beans and peas, peanuts, soybeans, dry beans, broad beans, dry peas, chickpeas, and lentils) represent an important component of the human diet in several areas of the world, especially in the developing countries, where they complement the lack of proteins from cereals, roots, and tubers. In some regions of the world, legume seeds are the only protein supply in the diet. The health benefits of legume consumption have received rising interest from researchers, and their consumption and production extends worldwide. Among European countries, higher legume consumption is observed around the Mediterranean, with per capita daily consumption between 8 and 23 g, while in Northern Europe, the daily consumption is less than 5 g per capita. The physiological effects of different legumes vary significantly. These differences may result from the polysaccharides composition, in particular, the quantity and variety of dietary fibers and starch, protein make-up, and variability in phytochemical content. The majority of legumes contain phytochemicals: bioactive compounds, including enzyme inhibitors, phytohemagglutinins (lectins), phytoestrogens, oligosaccharides, saponins, and phenolic compounds, which play metabolic roles in humans who frequently consume these foods. Dietary intake of phytochemicals may provide health benefits, protecting against numerous diseases or disorders, such as coronary heart disease, diabetes, high blood pressure and inflammation. The synergistic or antagonistic effects of these phytochemical mixtures from food legumes, their interaction with other components of the diet, and the mechanism of their action have remained a challenge with regard to understanding the role of phytochemicals in health and diseases. Their mitigating effects and the mechanism of their action need to be further addressed if we are to understand the role of phytochemicals in health and diseases. This review provides an overview of the nutritional quality of legumes and their potential contribution in cardiometabolic risk prevention.